Costs

Chapters 7

1 Costs that Matter for Decision Making: Opportunity Costs

Accounting Costs: The direct costs of operating a business, including costs for raw materials, wages paid to workers, rent paid for office or retail space, and the like.

Economic Costs: The sum of a producer's accounting and opportunity costs.

Opportunity Costs: The value of what a producer gives up by using an input.

Economic Profit: A firm's total revenue minus its economic cost.

Accounting Profit: A firm's total revenue minus its accounting cost.

Example: Cooke's Catering is owned by Dan Cooke. For the past year, Cooke's Catering had the following statement of revenues and costs:

Revenues	$500,000
Supplies	$150,000
Electricity and Water	$15,000
Employee Salaries	$50,000
Dan's Salary	$60,000

Dan has always had the option of closing his catering business and renting out his building for $100,000 per year. In addition, Dan currently has job offers from another catering company (offering a salary of $45,000 per year) and a high-end restaurant (at a salary of $75,000 per year). Dan can only work one job at any time.

(a) What is Cooke's Catering's accounting cost? 150,000+15,000+50,000+60,000=275,000

(b) What is Cooke's Catering's economic cost? 275,000+100,000+15,000=390,000

(c) What is Cooke's Catering's economic profit? 500,000-390,000=110,000. Dan should continue to operate his catering business.

2 Costs That Do Not Matter For Decision Making: Sunk Costs

> **Sunk Cost:** A cost that, once paid, cannot be recovered.

But suppose you are a "miner" of the cryptocurrency Bitcoin. Solving a particular type of math problem faster than other people do is how Bitcoin miners make money. There is a unique kind of computer chip known as an ASIC chip that is specially designed to solve the math problems involved in Bitcoin mining. This special design makes them far better at mining than a standard server's CPU. Because of their design, though, they are not very good at doing most other tasks people want computers to perform. They're just good at Bitcoin mining. So if you mine a bunch of bitcoins and then decide to close up your mining operation, the only likely buyer is another miner. If a number of miners decide to stop at once — and, with a 75% drop in Bitcoin prices in early 2018, hundreds of thousands did — the ASICs you paid so much for will no longer be worth much. The cost of those ASICs is a sunk cost; you cannot recover it even by shutting down.

> **Specific Capital:** Capital that cannot be used outside of its original application.

An important lesson about sunk costs is that once they are paid, they should not affect current and future production decisions.

The reason is simple: Because they are lost no matter what you choose to do, they cannot affect the relative costs and benefits of current and future production decisions.

Should a business stay open?

> **Operating Revenue:** The money a firm earns from selling its output.

> **Operating Costs:** The cost a firm incurs in producing its output.

Generally, a business should stay open if the value of staying open is greater than the value of shutting down. *Don't include the sunk costs when you add up those two sets of numbers. You will lose your sunk costs whether the business stays open or closes.*

> **Sunk Cost Fallacy:** The mistake of letting sunk costs affect forward-looking decisions.

POLL EVERYWHERE: A business anticipates future operating revenues of $200 million, future operating costs of $150 million, and sunk costs of $60 million. This business should:

3 Costs and Cost Curves

> **Variable Cost (TVC):** The cost of inputs that change as the firm changes its quantity of output.

> **Total Cost (TC):** The sum of a firm's fixed and variable costs: $TC = FC + TVC$.

3.1 Flexibility and Fixed versus Variable Costs

When a firm can easily adjust the levels of inputs it uses as output changes, the costs of these inputs are variable. When a firm cannot adjust how much of an input it buys as output varies, the input costs are fixed.

3.1.1 Time Horizon

Over very short time periods, many costs are fixed because a firm cannot adjust these input levels over short spans of time even if the firm's output level changes. As the time horizon lengthens, however, firms have greater abilities to change the levels of all inputs to accommodate output fluctuations. In the long run, all inputs are flexible.

Think about a restaurant. On a given day many costs are fixed (employees are scheduled and food inputs are ordered ahead of time). Over time costs become more variable.

3.1.2 Other Factors

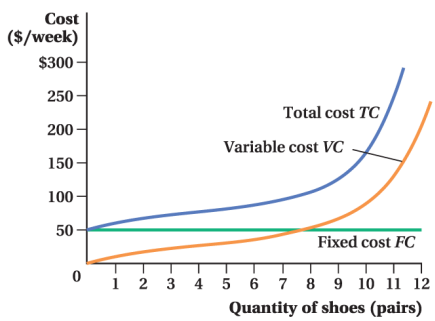
- The presence of active capital markets and resale markets. *These markets allow firms that need certain pieces of machinery or types of buildings only occasionally (like the construction firm that sometimes needs a crane to build houses) to pay for the input just when it's needed to make more output.*
- Labor contracts. *Some contracts require that workers be paid a specified amount regardless of how much time they spend producing output. These payments are fixed costs. For example, in the past, U.S. automakers' contracts included "jobs banks" for laid-off workers. These provisions required that laid-off workers receive 95% of their regular salary, retain their health benefits, and continue to accrue pension benefits until reaching retirement age.*

3.2 Deriving Cost Curves

> **Cost Curve:** The mathematical relationship between a firm's production costs and output.

Consider the example of Fleet Foot (FF), a running shoe company.

Output Quantity Q(Pairs of Shoes/Week)	Fixed Cost FC($/Week)	Variable Cost VC($/Week)	Total Cost TC($/Week)
0	50	0	50
1	50	10	60
2	50	17.5	67.5
3	50	22.5	72.5
4	50	25	75
5	50	30	80
6	50	37.5	87.5
7	50	47.5	97.5
8	50	60	110
9	50	75	125
10	50	100	150
11	50	150	200
12	50	225	275



Goolsbee et al., *Microeconomics*, 3e, © 2020 Worth Publishers

- Fixed cost does not vary with output, so it is constant and the fixed cost curve is horizontal.
- Variable cost changes with the output level. Fleet Foot's variable cost rises as output increases because FF must buy more inputs. The relationship between the amount of variable inputs a firm must buy and its output means the slope of the VC curve is always positive. The shape of FF's particular VC curve in Figure 7.1 indicates that the rate at which FF's variable cost increases first falls and then rises with output. Specifically, the curve becomes flatter as weekly output quantities rise from 0 to 4 pairs of shoes, indicating that the additional cost of producing another pair of shoes is falling as FF makes more shoes. However, at quantities above 4 pairs per week, the VC curve's slope becomes steeper. At these output levels, the additional cost of producing another pair of shoes is rising. We talk more about why this is the case later in the chapter.
- The total cost curve shows how a firm's total production cost changes with its level of output. Because all costs can be classified as either fixed or variable, the sum of these two components equals total cost. In fact, as is clear in Figure 7.1, the total and variable cost curves have the same shapes and are parallel to each other, separated at every point by the amount of fixed cost. Note also that when output is zero, total cost isn't zero; it is $50. Fleet Foot's fixed cost must be paid in the short run even when the firm produces no output.

4 Average and Marginal Costs

4.1 Average and Marginal Cost Measures

Average Fixed Cost: A firm's fixed cost per unit of output.

$$AFC = FC/Q$$

Average Variable Cost: A firm's per-unit variable cost of production.

$$AVC = TVC/Q$$

Average Total Cost: A firm's total cost per unit of output.

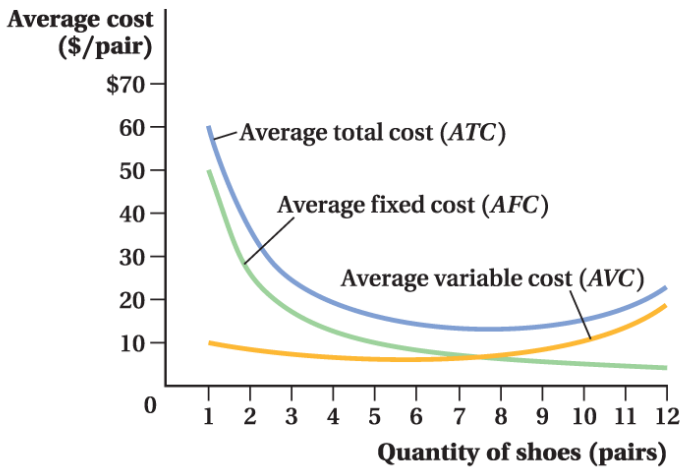
$$ATC = TC/Q = AFC + AVC$$

Marginal Cost: A firm's cost of producing one more unit of output.

$$MC = \Delta TC/\Delta Q = \Delta TVC/\Delta Q$$

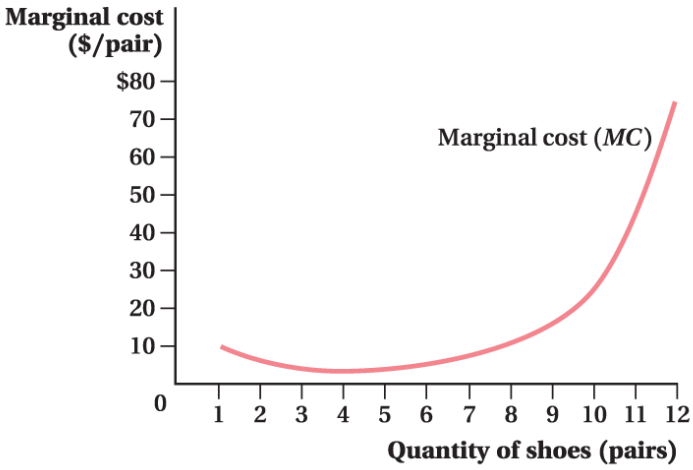
(1) Output Quantity Q	(2) Fixed Cost FC	(3) Variable Cost VC	(4) Total Cost TC	(5) Marginal Cost MC	(6) Average Fixed Cost FC/Q	(7) Average Variable Cost VC/Q	(8) Average Total Cost TC/Q
0	50	0	50	—	—	—	—
1	50	10	60	10	50	10	60
2	50	17.5	67.5	7.5	25	8.75	33.75
3	50	22.5	72.5	5	16.67	7.5	24.17
4	50	25	75	2.5	12.5	6.25	18.75
5	50	30	80	5	10	6	16
6	50	37.5	87.5	7.5	8.33	6.25	14.58
7	50	47.5	97.5	10	7.14	6.79	13.93
8	50	60	110	12.5	6.25	7.5	13.75
9	50	75	125	15	5.56	8.33	13.89
10	50	100	150	25	5	10	15
11	50	150	200	50	4.55	13.64	18.18
12	50	225	275	75	4.17	18.75	22.92

POLL EVERYWHERE: A firm is producing 50 units of output at a total cost of $1,000, with a per-unit variable cost of $8. What is the firm's average fixed cost?



Goolsbee et al., *Microeconomics*, 3e, © 2020 Worth Publishers

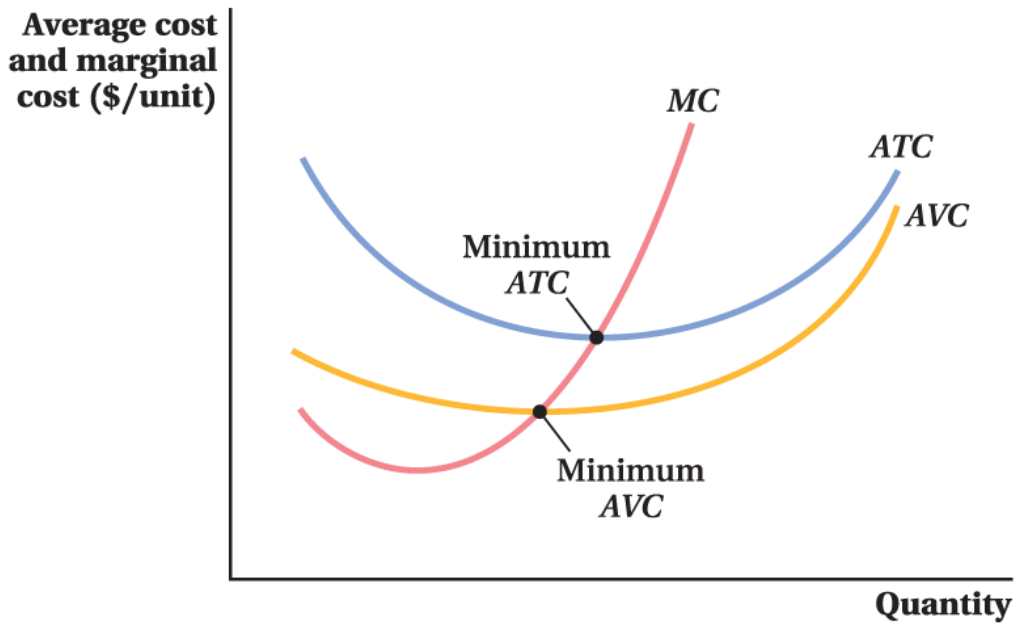
Fleet Foot's marginal cost initially declines as output increases. After a certain output level (4 units in Table 7.2), marginal cost begins to rise, and at even higher output levels, it rises steeply. Why? Marginal cost may initially fall at low quantities because of complications in producing the first few units that can be fixed fairly quickly, or because having more scale allows workers to specialize in those tasks that they are best at performing. As output continues to increase, however, these marginal cost reductions stop and marginal cost begins to increase with the quantity produced, as seen in Figure 7.3. There are many reasons why it becomes more and more expensive to make another unit as output rises: Capacity constraints may occur, inputs may become more expensive as the firm uses more of them, it may be more difficult for the company to coordinate its operations, and so on.



Goolsbee et al., *Microeconomics*, 3e, © 2020 Worth Publishers

Understanding the concept of marginal cost is critical: It is one of the most central concepts in all of economics, and it is the only cost that matters for many of the key decisions a firm makes.

4.2 Relationships between Average and Marginal Costs



Goolsbee et al., *Microeconomics*, 3e, © 2020 Worth Publishers

A couple of things to note about marginal and average cost:

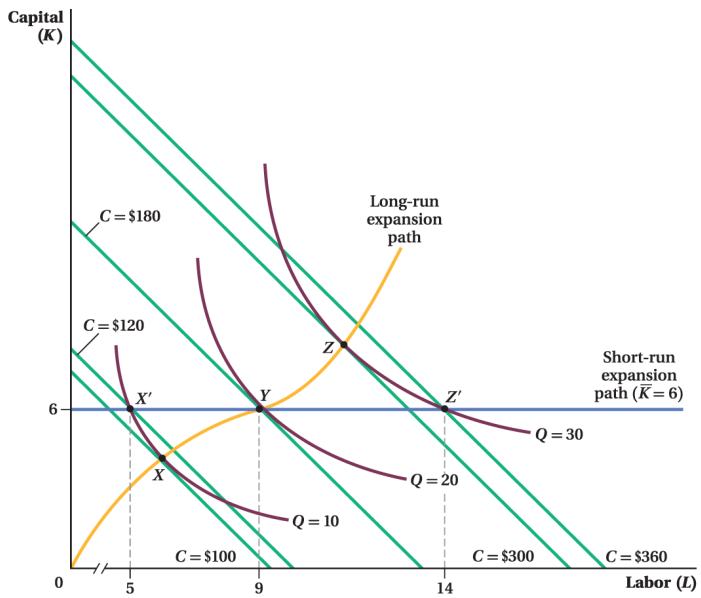
1. If the marginal cost of output is less than the average cost at a particular quantity, producing an additional unit will reduce the average cost because the extra unit's cost is less than the average cost of making all the units before it. *Suppose a firm is producing 9 units of output at an average cost of $100 per unit. If the marginal cost of the next unit is $90, the average cost will fall to* $(\$900 + \$90)/10 = \$99$.
2. When the marginal cost of the additional unit is above average cost, then producing it increases the average cost.
3. The marginal cost curve intersects the ATC and AVC curves at their lowest point.

5 Short-Run and Long-Run Cost Curves

5.1 Short-Run Production and Total Cost Curves

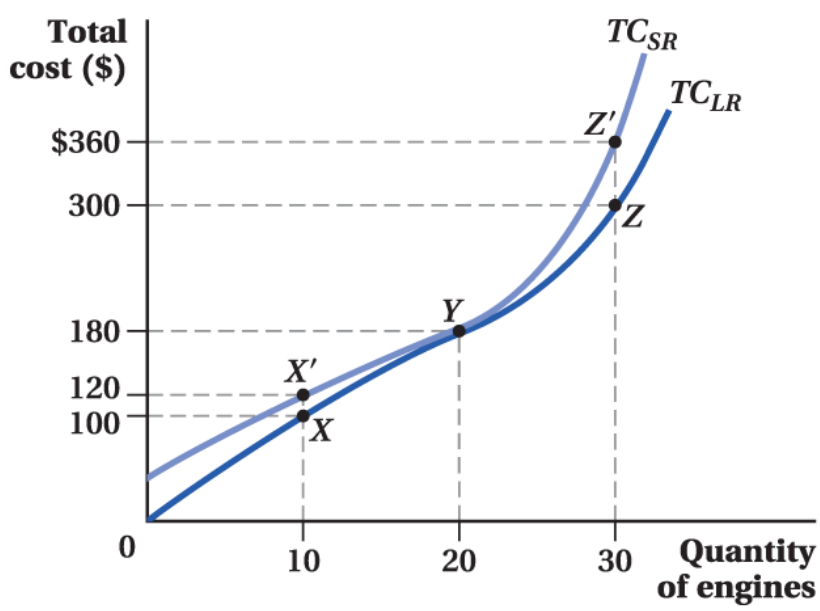
> **Short-Run Total Cost Curve:** The mathematical representation of a firm's total cost of producing different quantities of output at a fixed level of capital.

Consider the following firm. Suppose that initially the demand for this firm is 20 units, so that the firm chooses it's cost minimizing strategy of 6 units of capital and 9 units of labor. *This means that in the short-run, the firm will be stuck with 6 units of capital and must minimize their costs conditional on using 6 units of capital.*



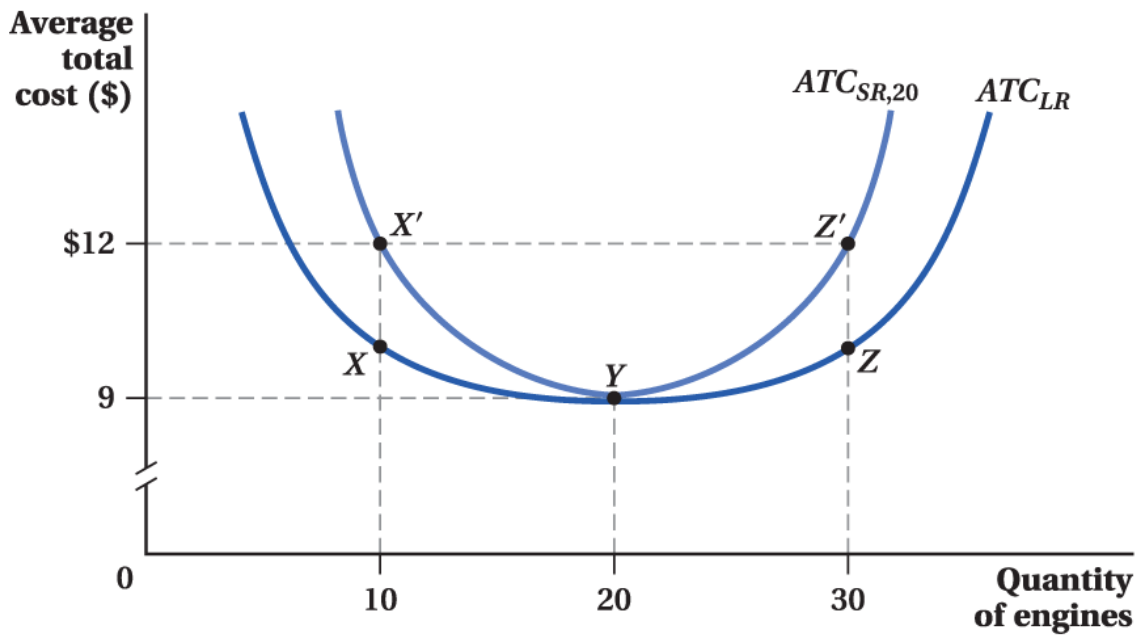
Goolsbee et al., *Microeconomics*, 3e, © 2020 Worth Publishers

We can plot out the short- and long-run total cost curves:



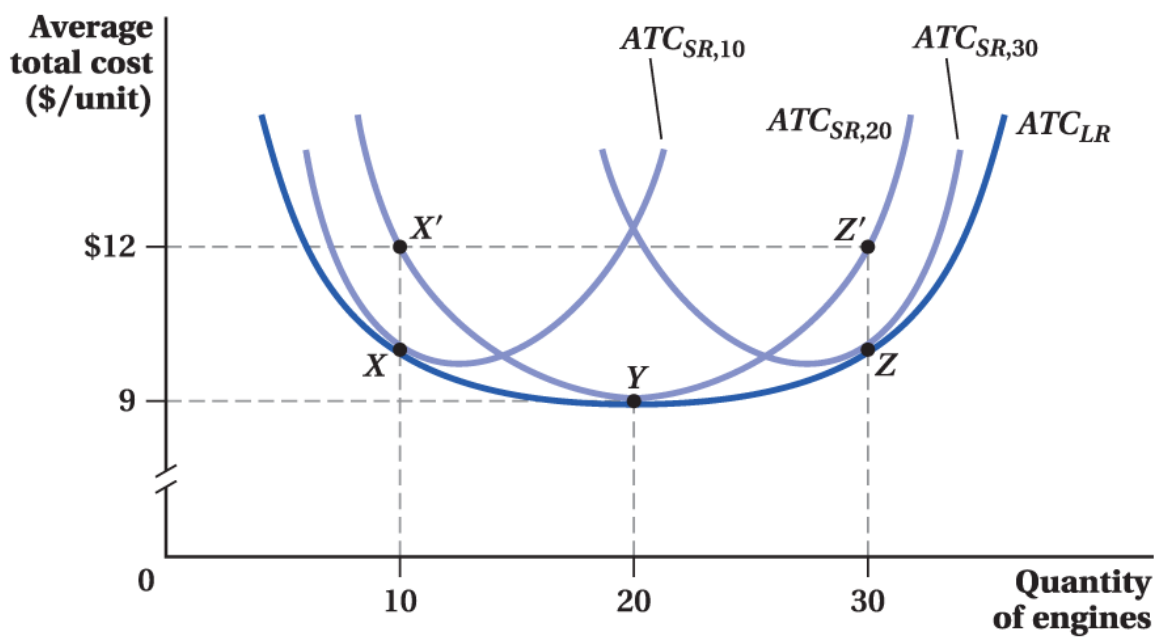
Goolsbee et al., *Microeconomics*, 3e, © 2020 Worth Publishers

5.2 Short-Run versus Long-Run Average Total Cost



Goolsbee et al., *Microeconomics*, 3e, © 2020 Worth Publishers

We've analyzed the distinction between long-run and short-run total and average total costs assuming that the short-run level of capital was fixed at 6 units (the level that minimized the cost of producing 20 engines). But suppose capital had been fixed at some other level instead. Say it was fixed at 9 units of capital, the level that minimizes the cost of producing 30 engines. (This is the capital level at point Z in Figure 7.6.)

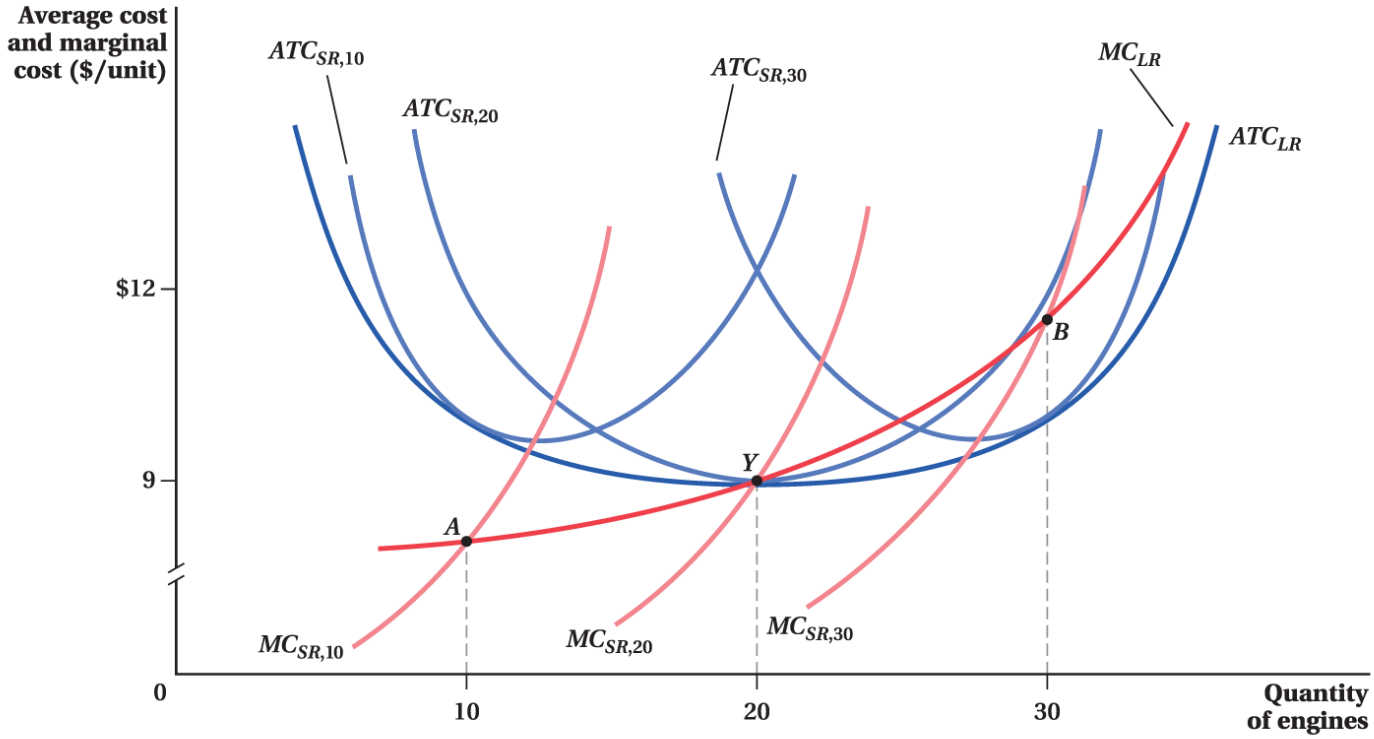


Goolsbee et al., *Microeconomics*, 3e, © 2020 Worth Publishers

One interesting thing to notice about this figure is that $ATC_{SR,10}$ and $ATC_{SR,30}$ do not touch ATC_{LR} at their lowest points. That's because even the output levels that minimize average total costs in the short run when capital is fixed (the low points on $ATC_{SR,10}$ and $ATC_{SR,30}$) can be produced more cheaply if capital inputs were flexible. In the one case where the short-run level of capital is fixed at the fully cost-minimizing level even if capital were flexible (the $ATC_{SR,20}$ curve), the two points are the same.

5.3 Short-Run versus Long-Run Marginal Cost Curves

Every short-run average total cost curve has a corresponding short-run marginal cost curve that shows how costly it is to build another unit of output when capital is fixed at some particular level. A short-run marginal cost curve always crosses its corresponding short-run average total cost curve at the minimum of the average total cost curve.



Goolsbee et al., *Microeconomics*, 3e, © 2020 Worth Publishers

To figure out the long-run marginal cost, we know that in the short run with capital fixed, there is only one output level at which a firm would choose that same level of capital: the output at which the short-run average total cost curve touches the long-run average total cost curve.

- If $Q = 10$, the business would choose a level of capital in the long-run ($K = 4$).
- Because the short- and long-run average total cost curves coincide at this quantity (but only at this quantity), so, too, do the short- and long-run marginal cost curves.
- Therefore, to find the long-run marginal cost of producing 10 units, we can go up to the short-run marginal cost curve at $Q = 10$.
- We can do the same for 20 and 30 units.
- When we connect these long-run marginal cost points A, Y, and B, along with the similar points corresponding to every other output quantity, we trace out the long-run marginal cost curve MCLR. Notice that, just as we have discussed, long-run marginal cost is below long-run average total cost when average total cost is falling (such as at point A), above long-run average total cost when average total cost is rising (such as at point B), and equal to long-run average total cost when average total cost is at its minimum (point Y).

6 Economies in the Production Process

6.1 Economics of Scale

> **Economies of Scale:** Total cost rises at a slower rate than output rises.

> **Diseconomies of Scale:** Total cost rises at a faster rate than output rises.

> **Constant Economies of Scale:** Total cost rises at the same rate as output rises.

- Because economies of scale imply that total cost increases less than proportionately with output, they also imply that long-run average total cost falls as output grows.
- Similarly, diseconomies of scale imply an upward-sloping long-run average total cost curve because total cost rises more quickly than output. Constant economies of scale make the long-run average total cost curve flat.
- Putting these together, we can see what the typical U-shaped, long-run average total cost curve implies about economies of scale.

POLL EVERYWHERE: Economies of scale occur when

6.2 Economies of Scope

> **Economies of Scope:** The simultaneous production of multiple products comes at a lower cost than if a firm made each product separately and then added up the costs.

$$SCOPE = \frac{TC(Q_1, 0) + TC(0, Q_2) - TC(Q_1, Q_2)}{TC(Q_1, Q_2)}$$

If $SCOPE > 0$, the total cost of producing Goods 1 and 2 jointly is less than making the goods separately, so there are economies of scope.

> **Diseconomies of Scope:** The simultaneous production of multiple products comes at a higher cost than if a firm made each product separately.

6.3 Where Economies of Scope Come From

- A common source of economies of scope is when different parts of a common input can be applied to the production of the firm's different products. *Cereal Production or oil refining (impossible to only make gasoline/kereosene/diesel/etc.)*

7 Finding a Firm's Total Cost Function from the Cost Minimization Problem

$$Q = K^{0.5} L^{0.5}$$

$$R = 1$$

$$w = 9$$

$$MP_L = 0.5 K^{0.5} L^{-0.5}$$

$$MP_K = 0.5 K^{-0.5} L^{0.5}$$

$$MRTS = \frac{MP_L}{MP_K} = \frac{0.5 K^{0.5} L^{-0.5}}{0.5 K^{-0.5} L^{0.5}} = \frac{K}{L}$$

$$MRTS = \frac{w}{R}$$

$$\frac{K}{L} = \frac{9}{1} \rightarrow K = 9L$$

$$Q = K^{0.5} L^{0.5}$$

$$Q = (9L)^{0.5} L^{0.5}$$

$$Q = 9^{0.5} L$$

$$Q = 3L$$

$$L = \frac{Q}{3}$$

$$K = 9 \times \frac{Q}{3}$$

$$K = 3Q$$

$$TC = wL + RK$$

$$TC = 9 \times \frac{Q}{3} + 1 \times 3Q$$

$$TC = 3Q + 3Q$$

$$TC = 6Q$$

Figure 1